THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Mangrove forests along coasts help to: 1

(a) stop wind **(b)** reduce rainfall
(c) increase cyclones **(d)** reduce the force of storm surges

2. A diver goes from 5 m to 10 m depth. The pressure on him: 1

(a) becomes zero **(b)** stays same **(c)** increases **(d)** decreases

3. When a syringe plunger is pulled out, liquid rises in it because: 1

(a) gravity decreases **(b)** the syringe attracts liquid
(c) liquid expands **(d)** atmospheric pressure pushes the liquid

4. A thunderstorm usually has: 1

(a) snow and hail only **(b)** dry weather
(c) lightning, thunder and heavy rain **(d)** calm winds only

5. In a container with water, the pressure at the same depth is: 1

(a) greater downward **(b)** zero sideways
(c) greater upward **(d)** equal in all directions

6. Which has the most pressure at the bottom of a tank of the same height filled with liquid? 1

(a) tank at any speed **(b)** the narrowest tank
(c) the widest tank **(d)** the liquid with the greatest density

7. A gentle wind blowing at 10 km/h is likely to be: 1

(a) a breeze **(b)** a cyclone **(c)** a hurricane **(d)** a tornado

8. The pressure at the bottom of a lake depends on: 1

(a) length of the lake **(b)** depth of water
(c) width of the lake **(d)** shape of the lake

9. A dropper works by: 1

(a) magnet
(b) gravity only
(c) squeezing and releasing the bulb to change air pressure
(d) friction

10. The red alert given by the weather department means: 1

(a) no change **(b)** extreme danger, take action
(c) all is well **(d)** light rain

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the kite fly better on a windy day? 2

12. Why is the pressure at the bottom of a tank more than at the top? 2

13. Why are houses on the coast built on stilts or raised platforms in cyclone-prone areas? 2

14. Why is it dangerous to take shelter under a tree during a thunderstorm? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the 'collapsing can' experiment. 3

16. Why does an inflated balloon burst when heated or pressed hard? 3

17. What is the role of ocean temperature in cyclone formation? Explain. 3

OR

Why do aircraft cabins need to be pressurised? 3

18. State three ways in which communities can reduce the harm from cyclones. 3

OR

Why is it dangerous to be near the coast during a cyclone's landfall? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how atmospheric pressure is shown by four simple activities. 5

OR

A cyclone is forecast to hit the east coast. List the preparations by the 5
(a) government
(b) family
(c) fisherman.

20. A water tank is filled with water to 3 m. Another tank has 6 m of water. Compare the pressures at the base. If a hole of the same size is opened at the base of each, compare the jets. 5

OR

Explain the nature of lightning, why it occurs in thunderstorms, and describe the working of a lightning conductor. 5

END OF QUESTION PAPER — Total: 40 Marks